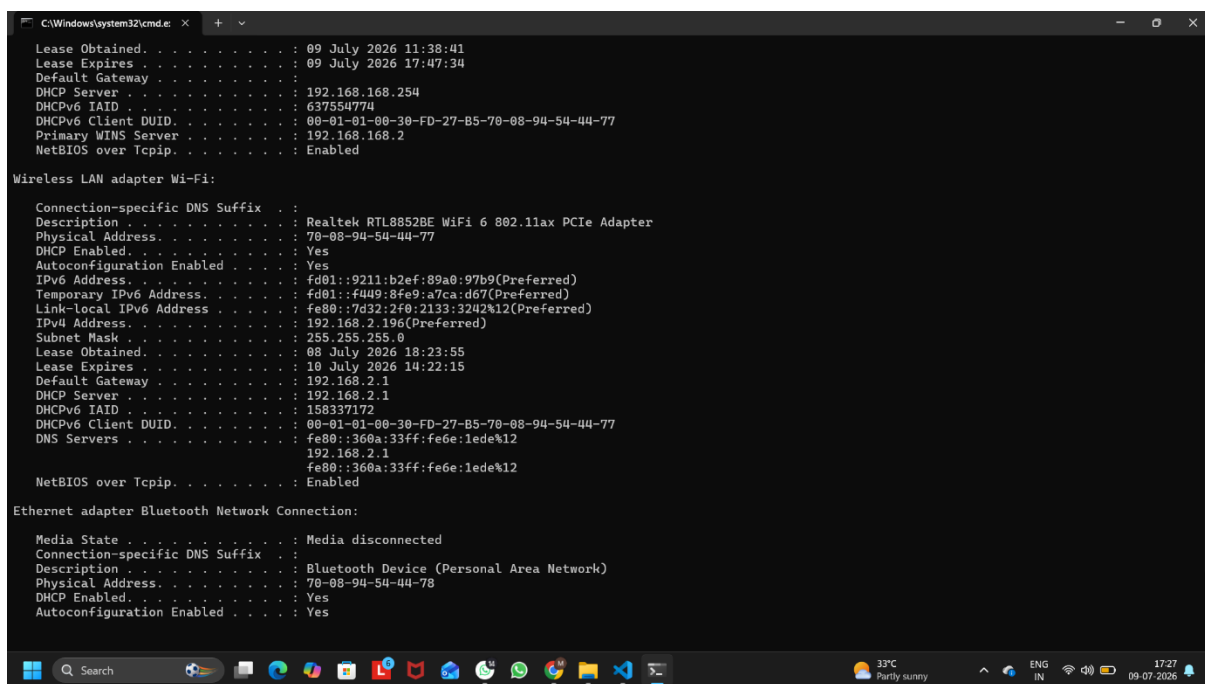


Network Troubleshooting, Management, Monitoring, and Optimization

Q1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Ans:

Figure 1: Output of the `ipconfig /all` Command



```
C:\Windows\system32\cmd.exe
Lease Obtained. . . . . : 09 July 2026 11:38:41
Lease Expires . . . . . : 09 July 2026 17:47:34
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.168.254
DHCPv6 IAID . . . . . : 637554774
DHCPv6 Client DUID. . . . . : 00-01-00-30-FD-27-B5-70-08-94-54-44-77
Primary WINS Server . . . . . : 192.168.168.2
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 78-08-94-54-44-77
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : fd01::9211:b2ef:89a0:97b9(Preferred)
Temporary IPv6 Address. . . . . : fd01::f449:8fe9:a7ca:d67(Preferred)
Link-local IPv6 Address . . . . . : fe80::7d32:2f0:2133:3242%12(Preferred)
IPv4 Address. . . . . : 192.168.2.196(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 08 July 2026 18:23:55
Lease Expires . . . . . : 10 July 2026 14:22:15
Default Gateway . . . . . : 192.168.2.1
DHCP Server . . . . . : 192.168.2.1
DHCPv6 IAID . . . . . : 158337172
DHCPv6 Client DUID. . . . . : 00-01-01-00-30-FD-27-B5-70-08-94-54-44-77
DNS Servers . . . . . : fe80::360a:33ff:fe6e:1ede%12
                        192.168.2.1
                        fe80::360a:33ff:fe6e:1ede%12
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 78-08-94-54-44-78
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

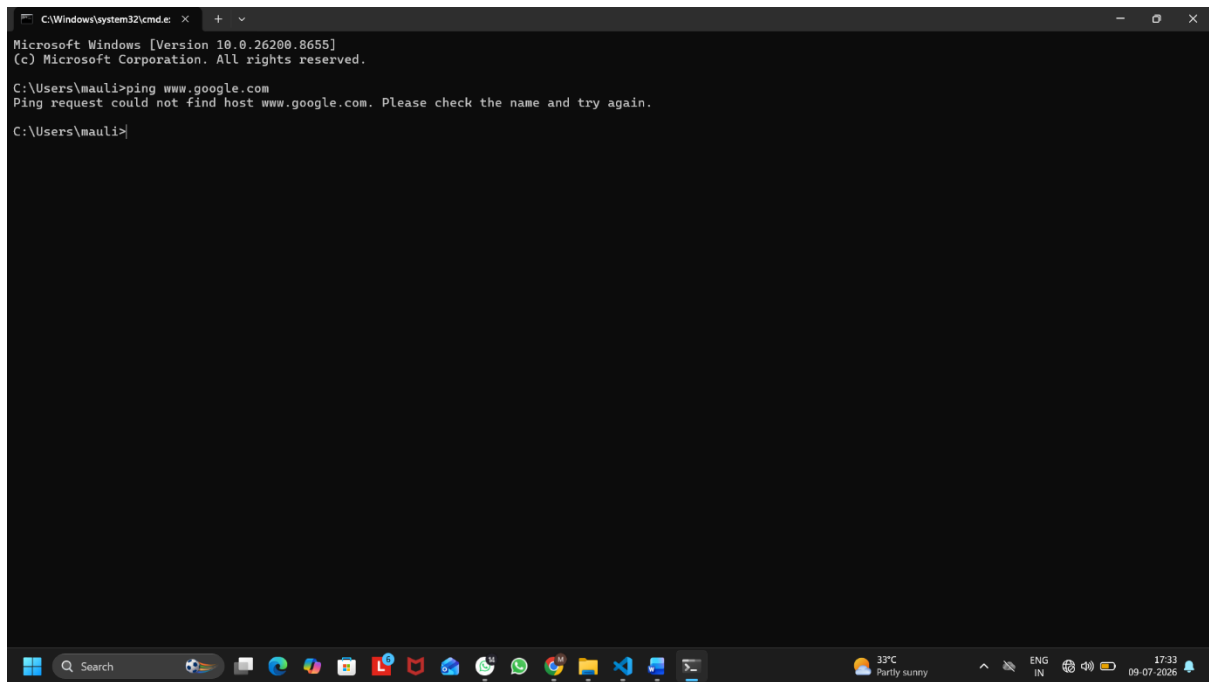
Network Configuration:

- IPv4 Address: 192.168.2.196
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1
- DNS Server(s):
 - fe80::360a:33ff:fe6e:1ede%12
 - 192.168.2.1

Q2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`.

Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Ans: Figure 1: Ping Result with Internet Disconnected



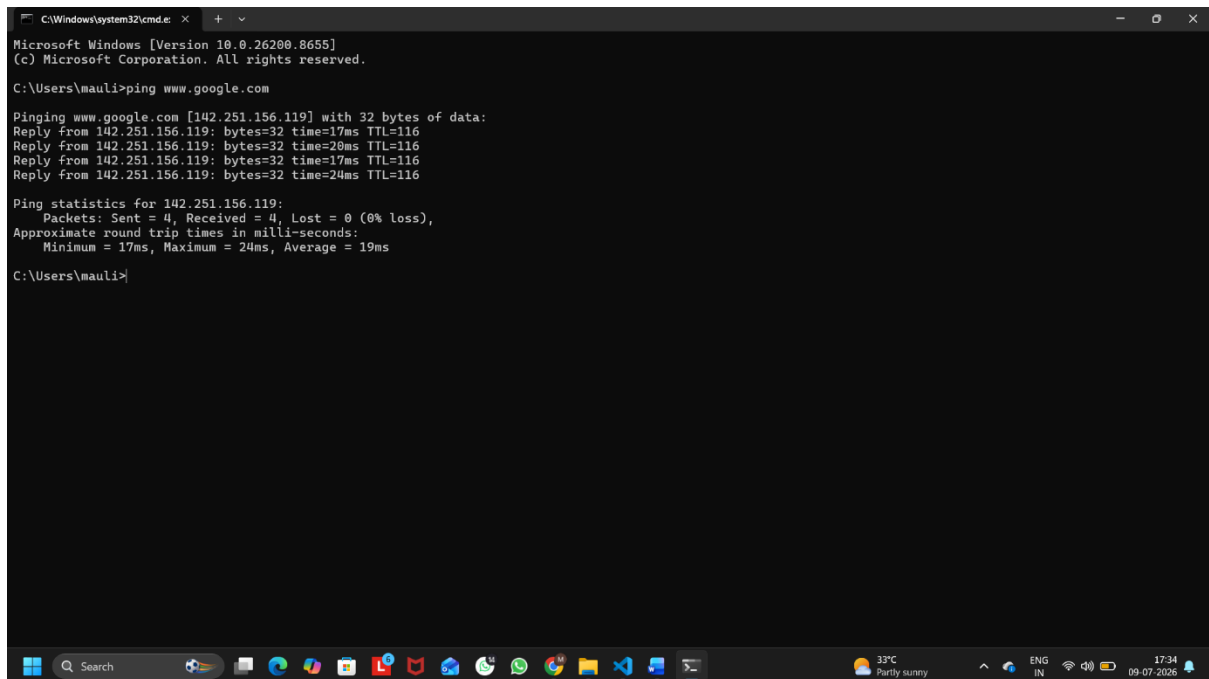
```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mauli>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\mauli>
```

The screenshot shows a Windows Command Prompt window. The title bar reads "C:\Windows\system32\cmd.exe". The window content shows the output of a ping command to www.google.com. The message "Ping request could not find host www.google.com. Please check the name and try again." is displayed, indicating that the internet is disconnected. The Windows taskbar at the bottom shows the system clock as 17:33 on 09-07-2026.

Figure 2: Ping Result with Internet Connected



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mauli>ping www.google.com

Pinging www.google.com [142.251.156.119] with 32 bytes of data:
Reply from 142.251.156.119: bytes=32 time=17ms TTL=116
Reply from 142.251.156.119: bytes=32 time=20ms TTL=116
Reply from 142.251.156.119: bytes=32 time=17ms TTL=116
Reply from 142.251.156.119: bytes=32 time=24ms TTL=116

Ping statistics for 142.251.156.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 24ms, Average = 19ms

C:\Users\mauli>
```

The screenshot shows a Windows Command Prompt window. The title bar reads "C:\Windows\system32\cmd.exe". The window content shows the output of a ping command to www.google.com. The output displays four successful replies from 142.251.156.119 with varying round trip times (17ms, 20ms, 17ms, 24ms) and a TTL of 116. Ping statistics show 4 packets sent and received with 0% loss. The Windows taskbar at the bottom shows the system clock as 17:34 on 09-07-2026.

Observation

Before Reconnecting (Internet Disconnected):

- The ping command failed and no reply was received from Google.
- This happened because the computer was not connected to the internet.

After Reconnecting (Internet Connected):

- The ping command successfully received replies from Google.
- This indicates that the internet connection was restored and the network is working properly.

Difference in Results:

Internet Disconnected	Internet Connected
No response from Google.	Reply received from Google.
Connection failed.	Connection successful.
Network or internet unavailable.	Network and internet working properly.

What Each Status Means:**1. Ping request could not find host www.google.com**

- This status indicates that the computer could not resolve the website name.
- It usually occurs when the internet connection is unavailable or DNS services are not working properly.

2. Reply from 142.251.156.119

- This status indicates that the destination server (Google) is reachable.
- The network connection is working properly and data packets are successfully sent and received.

3. Time = 17ms, 20ms, etc.

- It shows the time taken for a packet to travel to the destination and return.
- Lower values indicate better network performance.

4. TTL = 116

- TTL (Time To Live) indicates the maximum number of network hops a packet can travel before being discarded.
- It helps prevent packets from circulating indefinitely in the network.

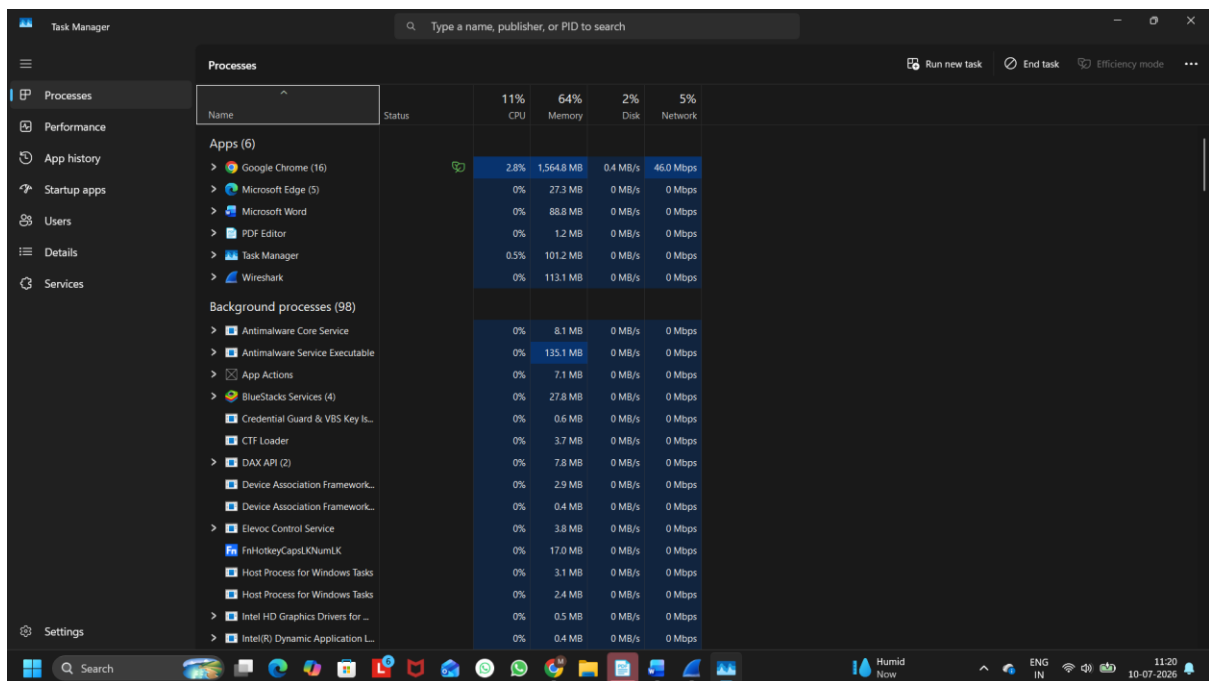
5. Lost = 0 (0% loss)

- This indicates that no packets were lost during transmission.
- A 0% packet loss means the network connection is stable and reliable.

Q3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Ans:

Figure 1: Network Usage Monitored in Task Manager While Streaming a YouTube Video



Peak Network Usage: **46.0 Mbps**

Process Using the Most Bandwidth: **Google Chrome (chrome.exe)**

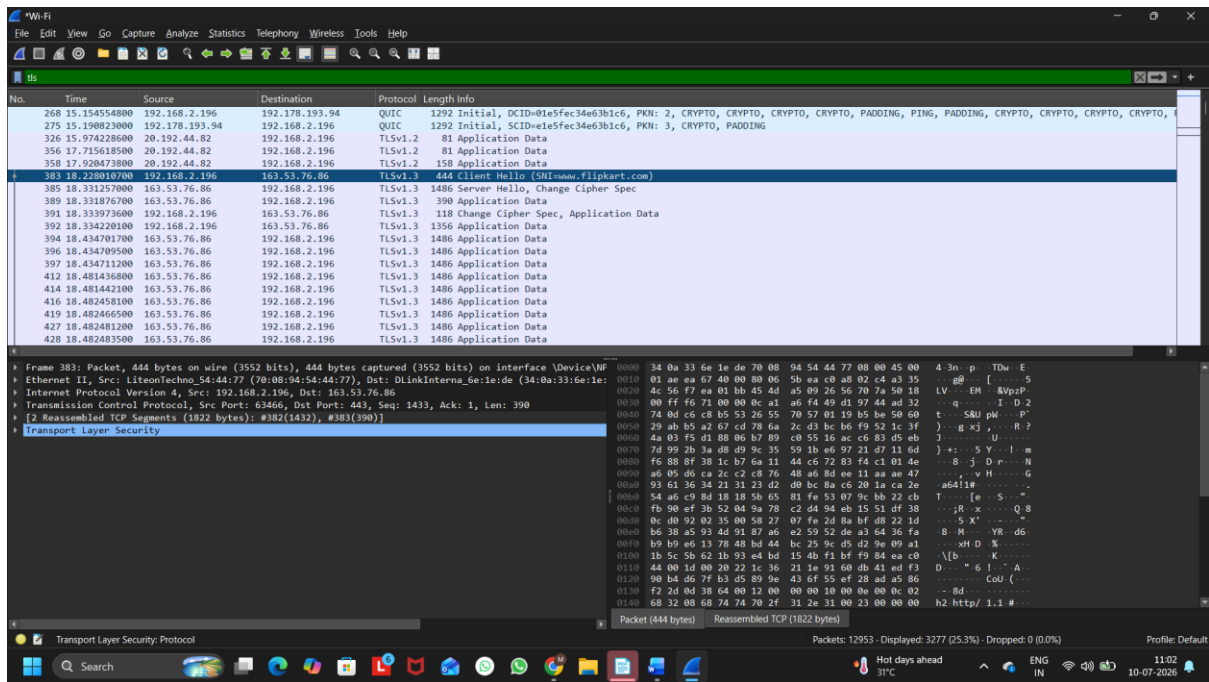
Observation:

I monitored the network usage for 5 minutes while streaming a YouTube video using Google Chrome. The highest network usage recorded was 46.0 Mbps. During this time, Google Chrome used the most bandwidth because it was streaming video content over the internet.

Q4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.
Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

Ans:

Figure 1: HTTPS (TLS) Traffic Captured Using Wireshark While Browsing Flipkart



Three Types of Information Visible in the Filtered Packets:

1. Source IP Address

- Shows the IP address of the sender (e.g., 192.168.2.196).

2. Destination IP Address

- Shows the IP address of the destination server (e.g., 163.53.76.86).

3. Protocol

- Shows the protocol being used for secure communication, such as TLSv1.3 (HTTPS).

Q5. Imagine your friend is complaining about slow internet while gaming online.

Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Ans:

AI-Generated Troubleshooting Checklist for Slow Internet While Gaming:

- Restart the modem and router to refresh the internet connection.
- Use a wired Ethernet connection instead of Wi-Fi for a more stable and faster connection.
- Close unnecessary apps and downloads that may be using internet bandwidth.
- Place the Wi-Fi router in a central location to improve signal strength and reduce interference.
- Update the router firmware and network drivers to ensure better performance and security.

Suggestion I Would Try First:

I would first restart the modem and router because it is a simple and quick solution that often resolves temporary network issues and improves internet performance.